Comprehensive database of sound changes reveals tree-like and wave-like processes in the Mixtec language family

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The Mixtec (Tu'un Savi) languages of southern Mexico

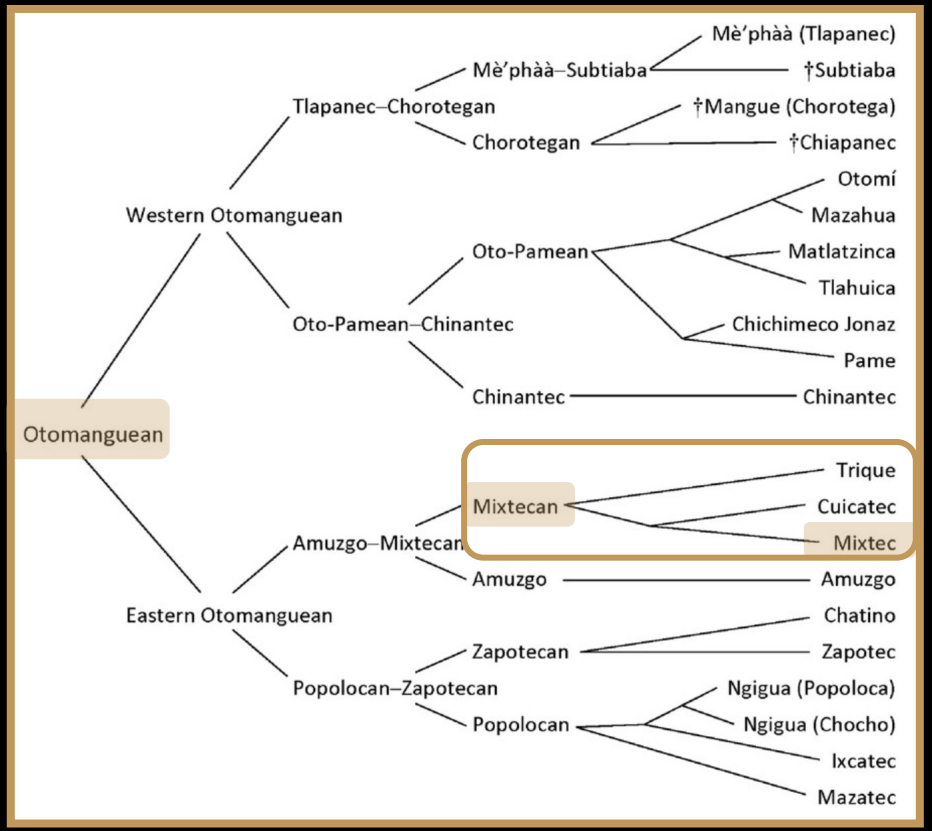
Geography



Mapa 1. Las divisiones coloniales y contemporáneas del Ñuu Savi*.

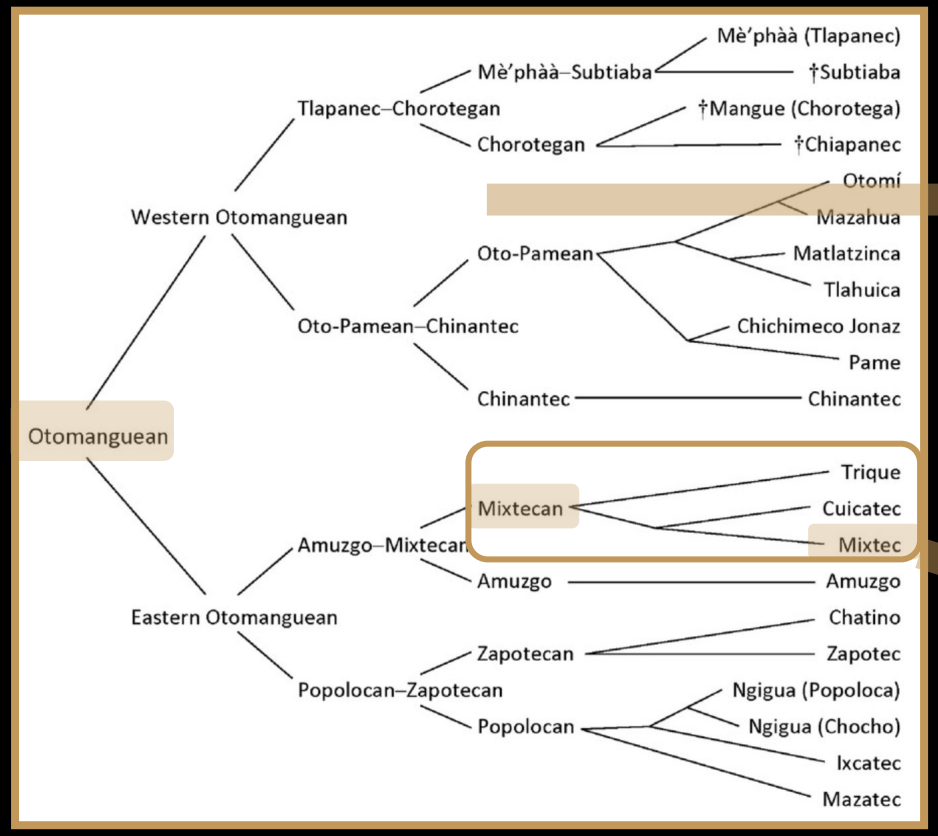
Aguilar Sánchez 2020:24. *Ñuu Savi: Pasado, presente y futuro.* Diss U Leiden

Genealogy



Campbell 2017 *Lang. Linguist. Compass 11(7)*
(based on Kaufman 2006, 1988)

Mixtec (Tu'un Savi) classification





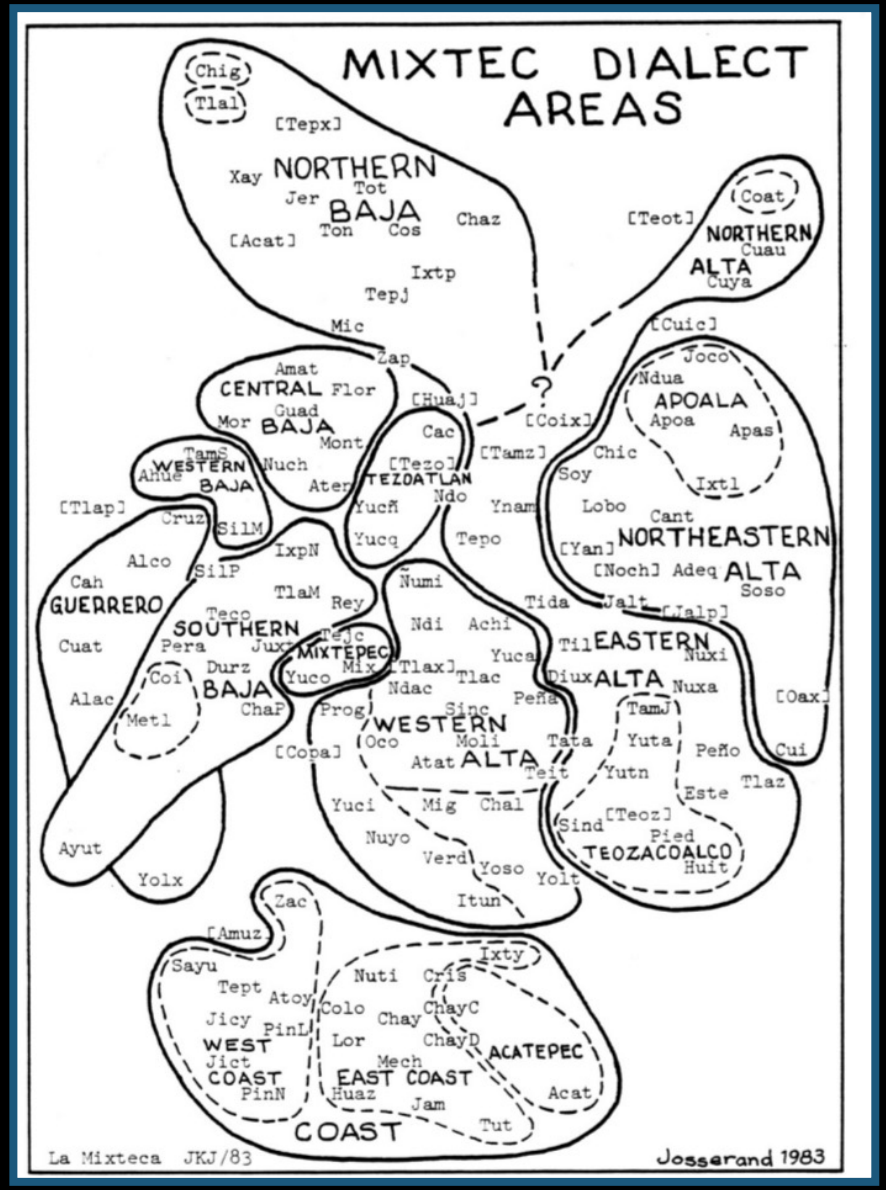


Campbell 2017. *Lang. Linguist. Compass 11(7)*
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The dialect area ‘problem’

“El problema es la aparente violación o infracción de las expectativas ideales del método comparativo en la linguística histórica (…). Pero este modelo presupone una efectiva separación de los grupos, después de un cambio que los divide. En contraste, en la Mixteca (…) vemos una situación más dinámica en el desarrollo de las familias lingüísticas.”

(Bradley & Josserand 1982: 303. *Anales de Antropol.* 19(2))

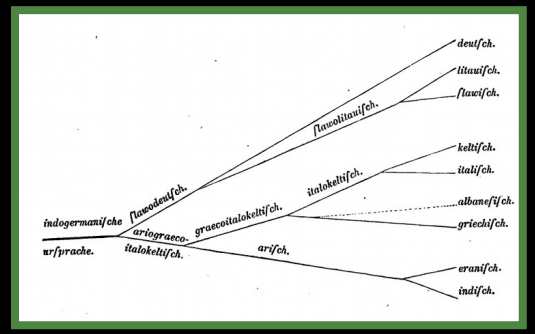


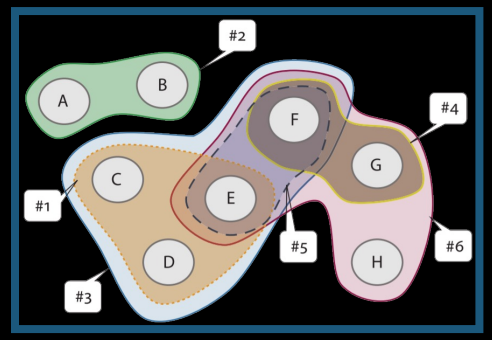
Josserand 1983. *Mixtec dialect history.* Diss. Tulane U

Tackling sound change in Mixtec

comparative method:

- regular sound change
- language relationships & subgrouping
- inheritance vs. contact





`dialect continuum/area/chains':

- wave-like (not tree-like) patterns
- contact among related languages
- challenges to applying the comparative method (Ross & Durie 1996, Kalyan & François 2018)

sound change in Mixtec?

Kalyan & François 2018. *Senri Ethn. St.* 98 | Ross & Durie 1996. *The Comp. Method Reviewed.* OUP
Image sources: https://de.m.wikipedia.org/wiki/Datei:Schleicher_Tree.jpg | Kalyan et al. 2019:2. *JoHL* 9(1)

Tackling sound change in Mixtec

Proto-Mixtec reconstructions and sound changes:

- Mak & Longacre 1960
- Bradley & Josserand 1982
- Josserand 1983
- Dürr 1987
- Swanton & Mendoza 2021
- Swanton 2021

Tackling sound change in Mixtec

Proto-Mixtec reconstructions and sound changes:

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Goal:

compile comprehensive, bottom-up data base of Mixtec sound change

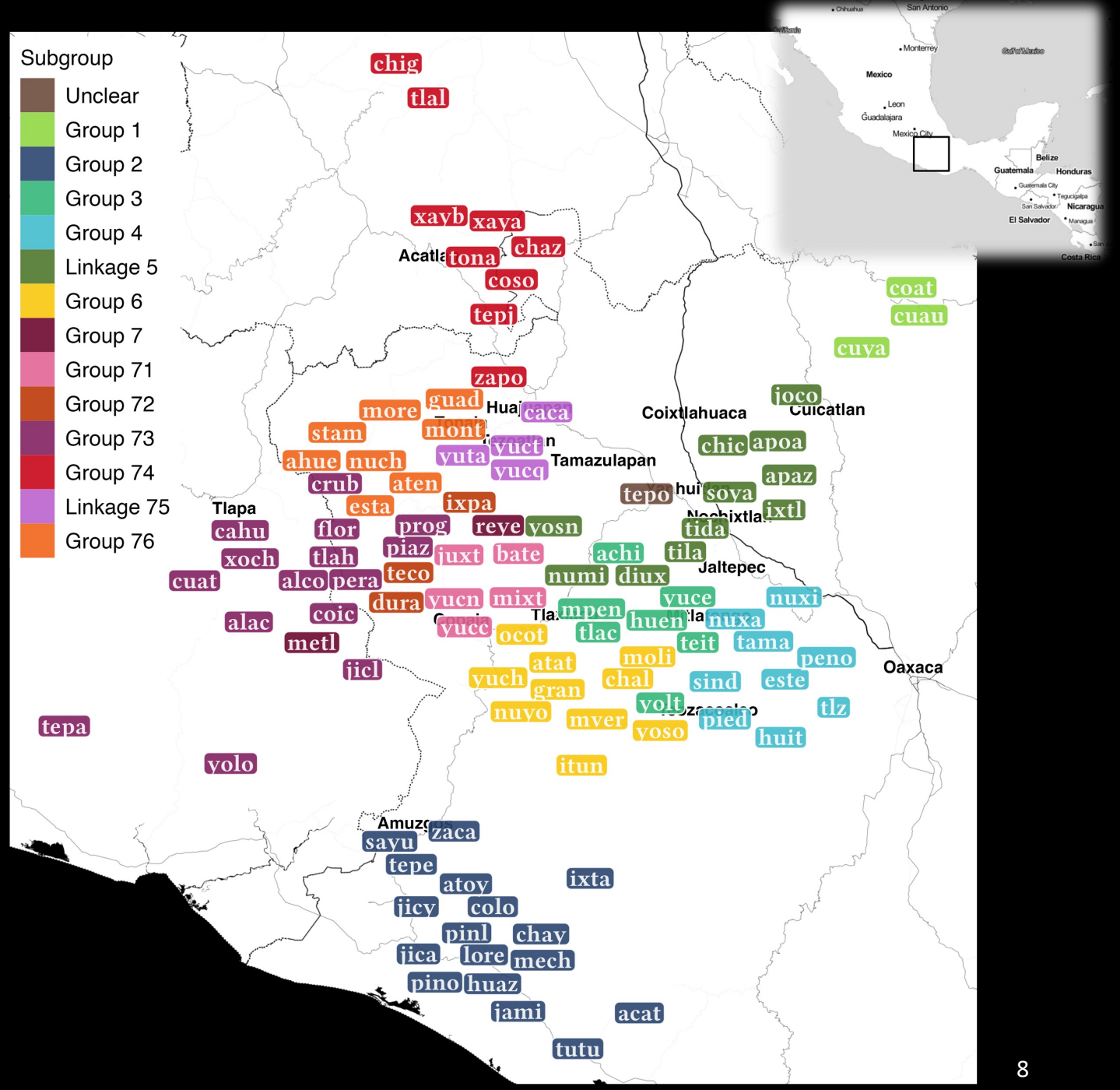
to

investigate the patterns and distributions

Sampled languages

105 varieties

covering all subgroups*



*Auderset et al. 2023. *Journal of Lang. Evolution*

Workflow – data base construction

AUTOTYP principles:

- autotypology
- modularity & connectivity
- separation of definition & coding files
- late aggregation

Witzlack-Makarevich et al. 2022. in: *Open Handbook of Ling Data Managment.* MIT

Workflow

input: subset of annotated cognate sets in IPA from Auderset et al. 2023

ID	MEANING	DOCULECT	SOURCE_ORTHOGRAPHIC	TOKENS	COGID
2105	SALT	SanJuanDiuxiMixtec	*ñɨ̃ɨ̃*	ɲ i ⁵ i ⁵	624
20187	SALT	SanMartinDuraznosMixtec	*ñìì*	ɲ i ¹ i ¹	624
6646	SALT	SantaMariaZacatepecMixtec	*ñii*	ɲ i i	624
2127	RIVER	SanJuanDiuxiMixtec	*yúté*	ʒ u ⁵ t e ⁵	607
19559	RIVER	SanMartinDuraznosMixtec	*yìtxa*	ʒ i ¹ tɕ a ³	607
12814	RIVER	SantaMariaZacatepecMixtec	*ju²ʧa²¹*	j u ³ tʃ a ³¹	607

cognates sets

AUTOTYP principles:

- autotypology
- modularity & connectivity
- separation of definition & coding files
- late aggregation

establish sound correspondences, propose/revise proto-forms

COGID	MEANING	PMX	PMX_Josserand	JosserandID	PMX_Durr	DurrID
294	GRASS	**ite*				
662	SLOW	**kʷeji*	kʷeje	163		
607	RIVER	**jute*	jute	23	jute	58
107	CHILI PEPPER	**jaʔaʔ*			jaʔaʔ	12
624	SALT	**jɨ̃ɨ̃ʔ*	jɨ̃ʔ	41		

proto-forms

Workflow

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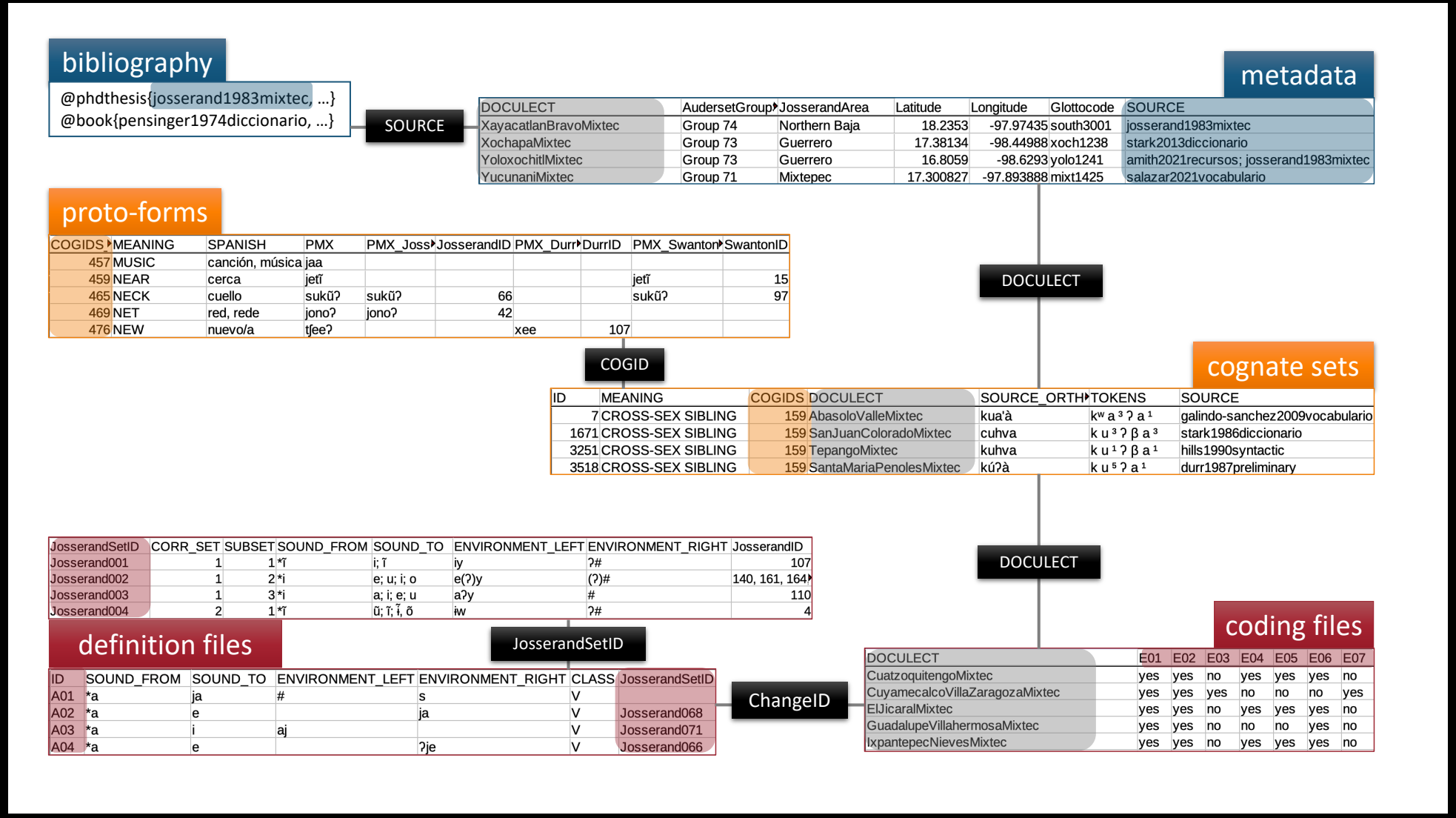
coding file

DOCULECT	E17	U32	Y01	J03	T12	T14
SanJuanDiuxiMixtec	no	no	no	yes	no	no
SanMartinDuraznosMixtec	yes	yes	yes	yes	yes	no
SantaMariaZacatepecMixtec	yes	no	yes	yes	no	yes

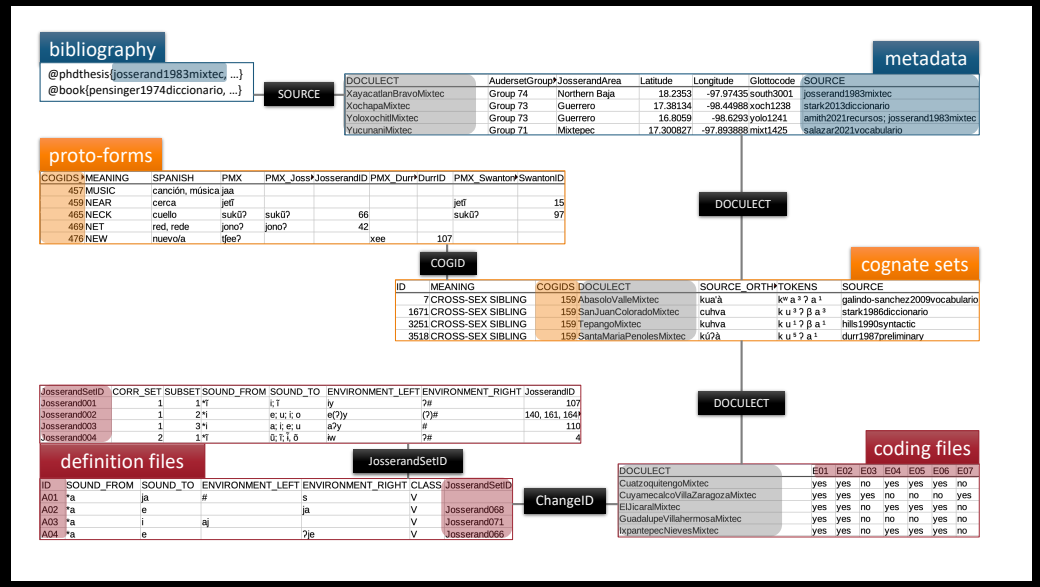
definition file

ID	SOUND_FROM	SOUND_TO	ENVIRONMENT_LEFT	ENVIRONMENT_RIGHT
E17	*e	a	i,ut	
U32	*u	i	j	te
Y01	*ɨ	i		(ʔ)#
J03	*j	ɲ	#	V(ʔ)Ṽ
T12	*t	tɕ		{ɨ,e}
T14	*t	tʃ		e

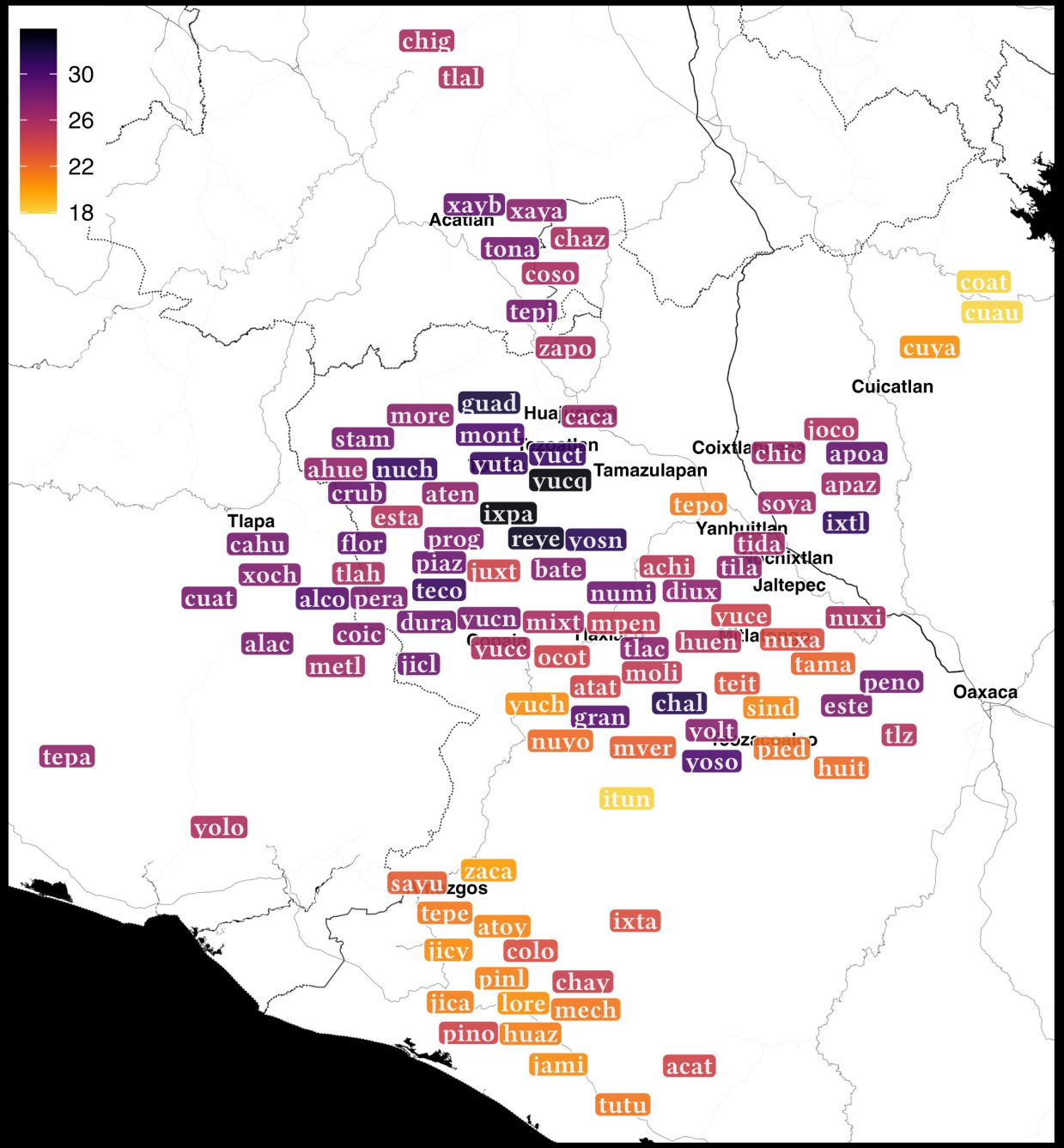
Structure of the data base



Current state of the data base



- 105 varieties
- 256 reconstructed proto-forms
- > 14’000 lexical entries
- 245 sound changes identified
  - 227 used in analysis:
    - 110 vowel changes
    - 117 consonant changes
- >25’000 data points (= coded sound changes)

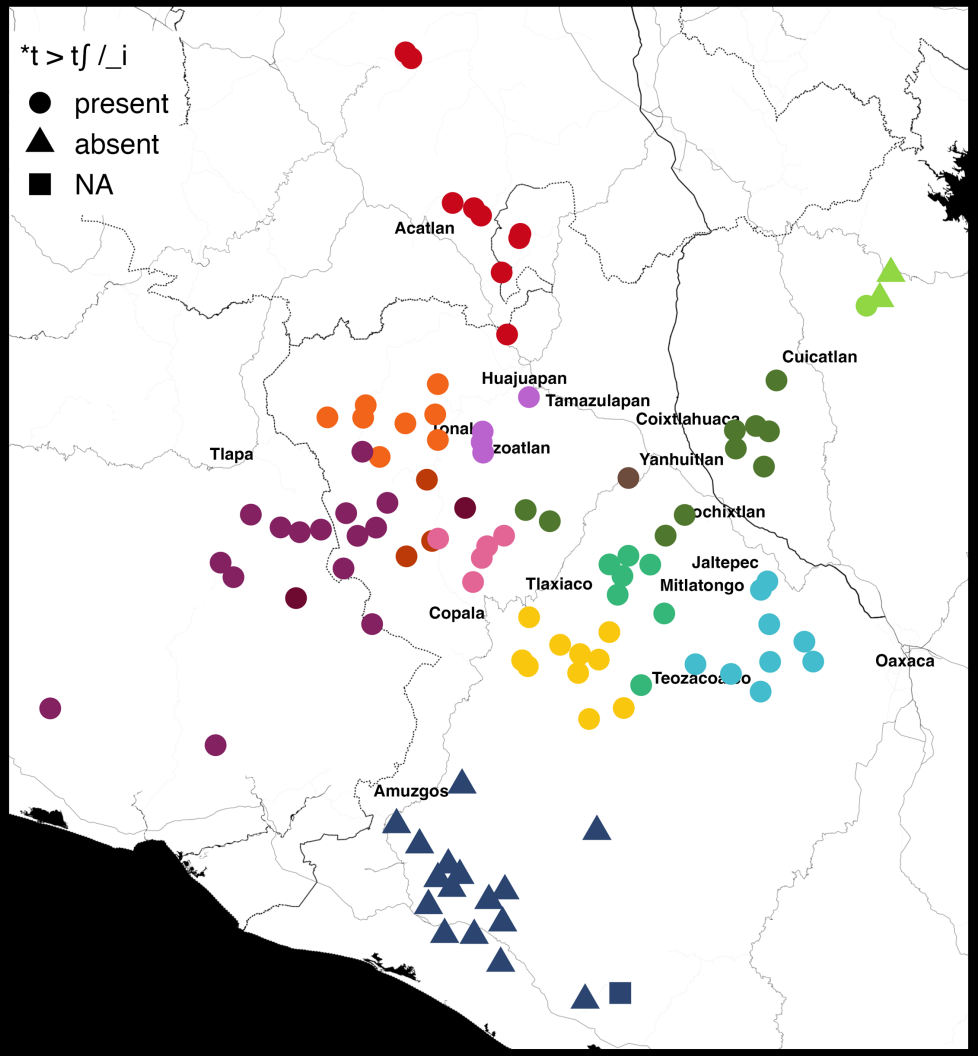


Wave-like processes

Map displaying the percentage of changes present per variety calculated as:

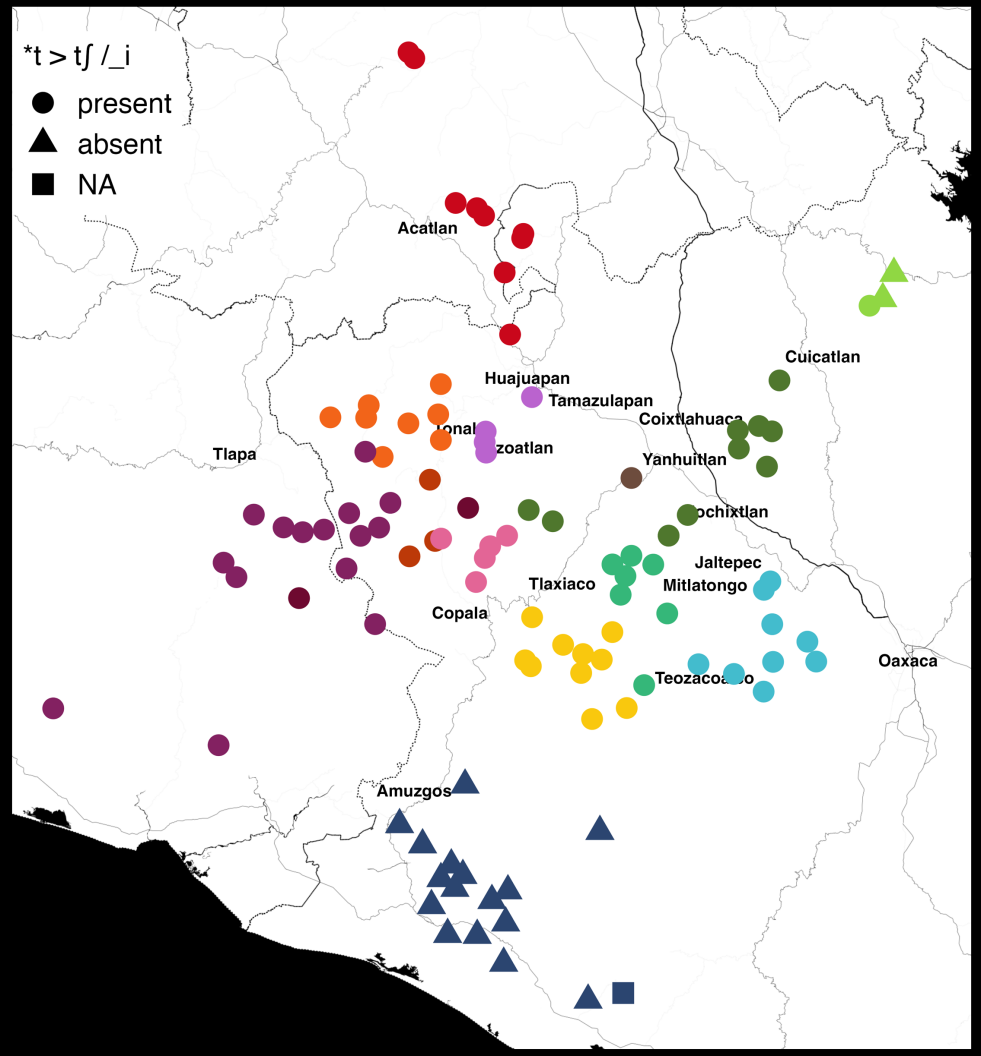
$$\frac{\text{(total number of changes) – (NA)}}{\text{(changes present)}}$$

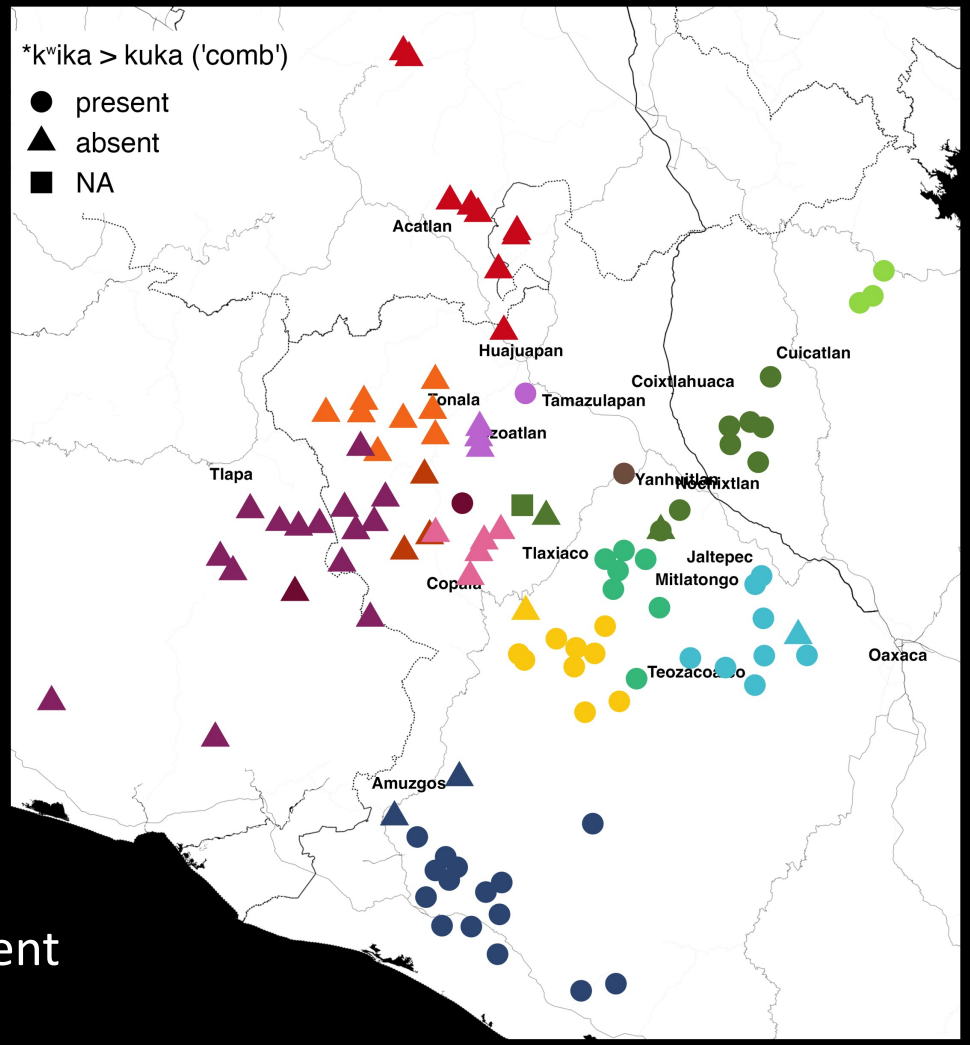
Tree-like processes



colors represent subgroups

Tree-like processes

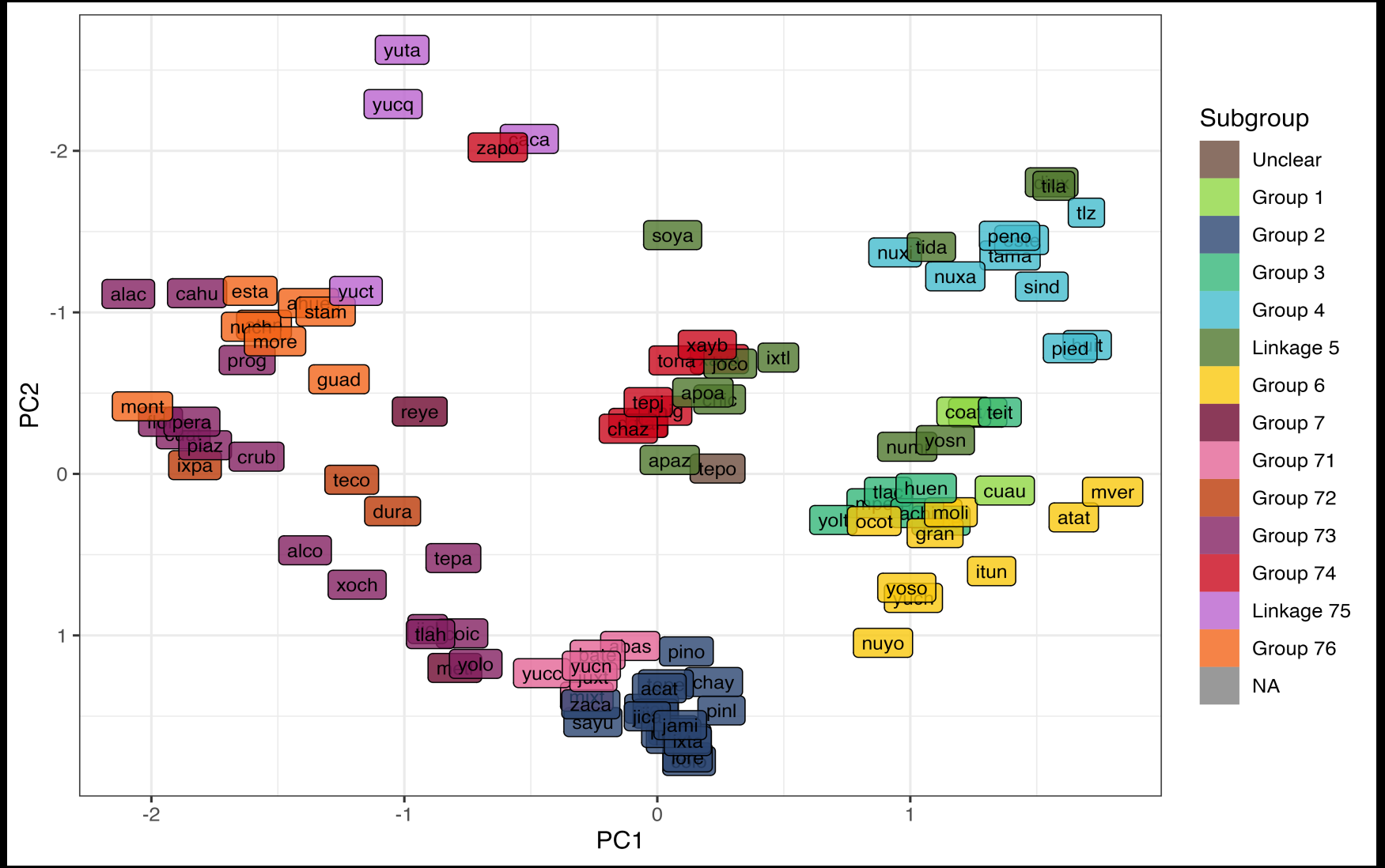




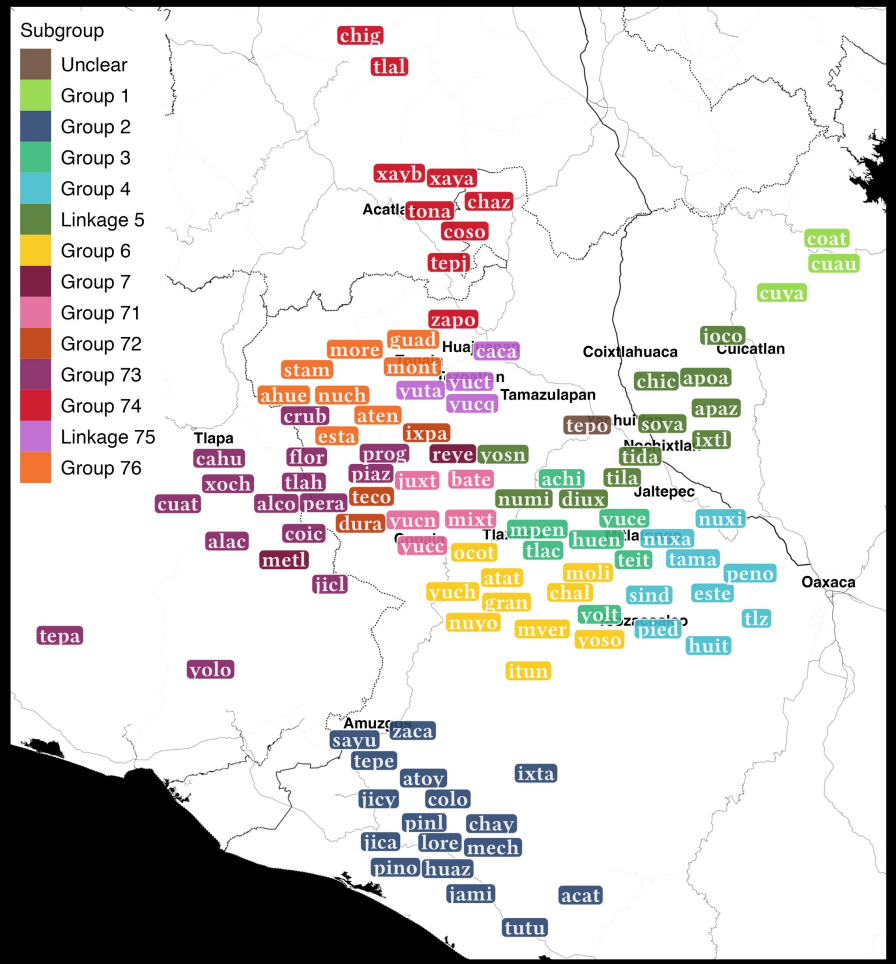
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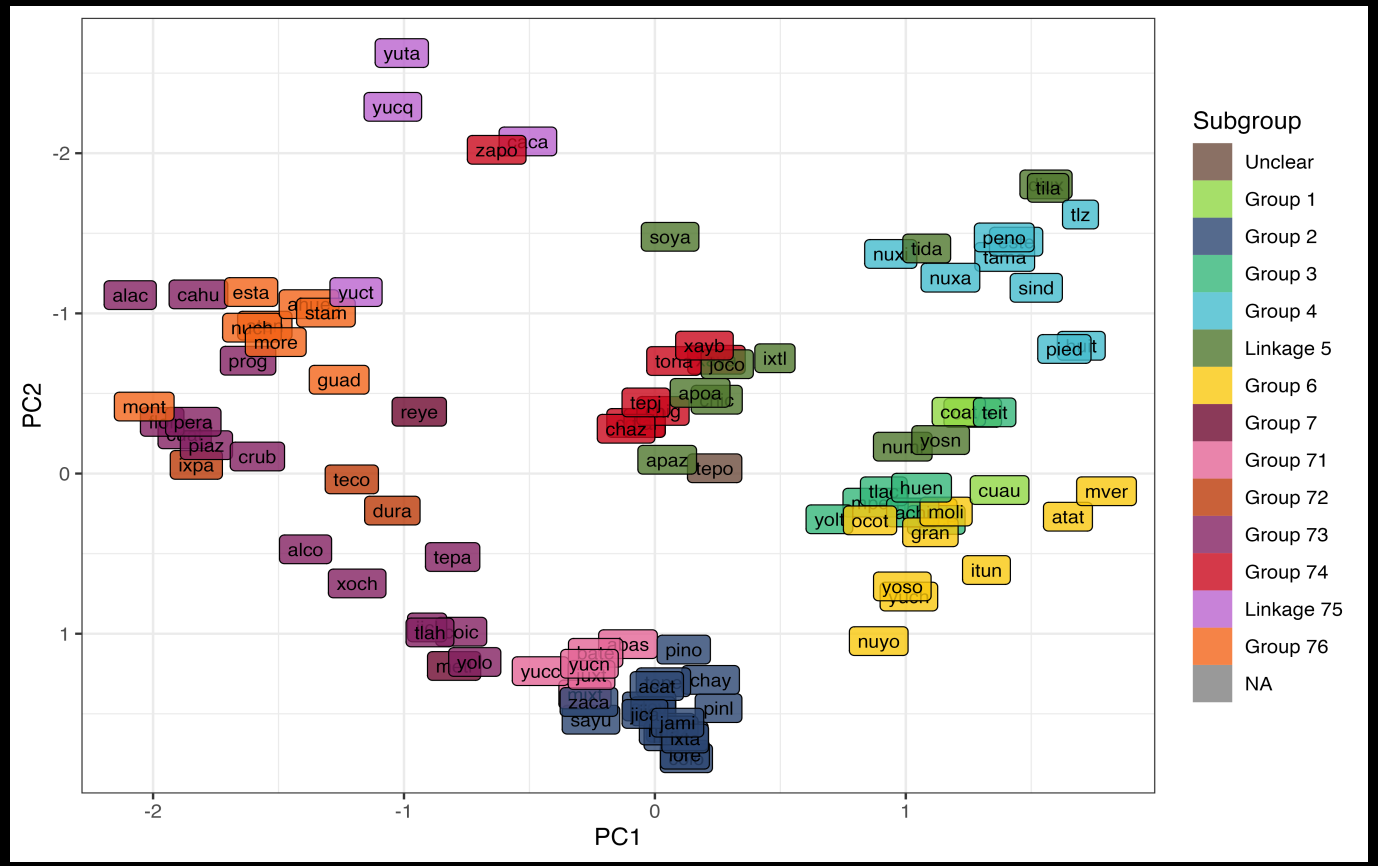
Beyond trees and waves: PCA

```
bpca calculated PCA
Importance of component(s):
                  PC1    PC2
R2             0.1807 0.1205
Cumulative R2 0.1807 0.3011
227     Variables
105     Samples
1431    NAs ( 6.004 %)
```



Beyond trees and waves





comprehensive database of sound changes coded in a bottom-up, fine-grained way reveals both tree-like and wave-like processes in the Mixtec language family

Limitations and future work:

- tone change
- relative chronology
- correlations between changes ~ chain shifts
- generalizations
- revision and expansion of the data base with new cognate sets

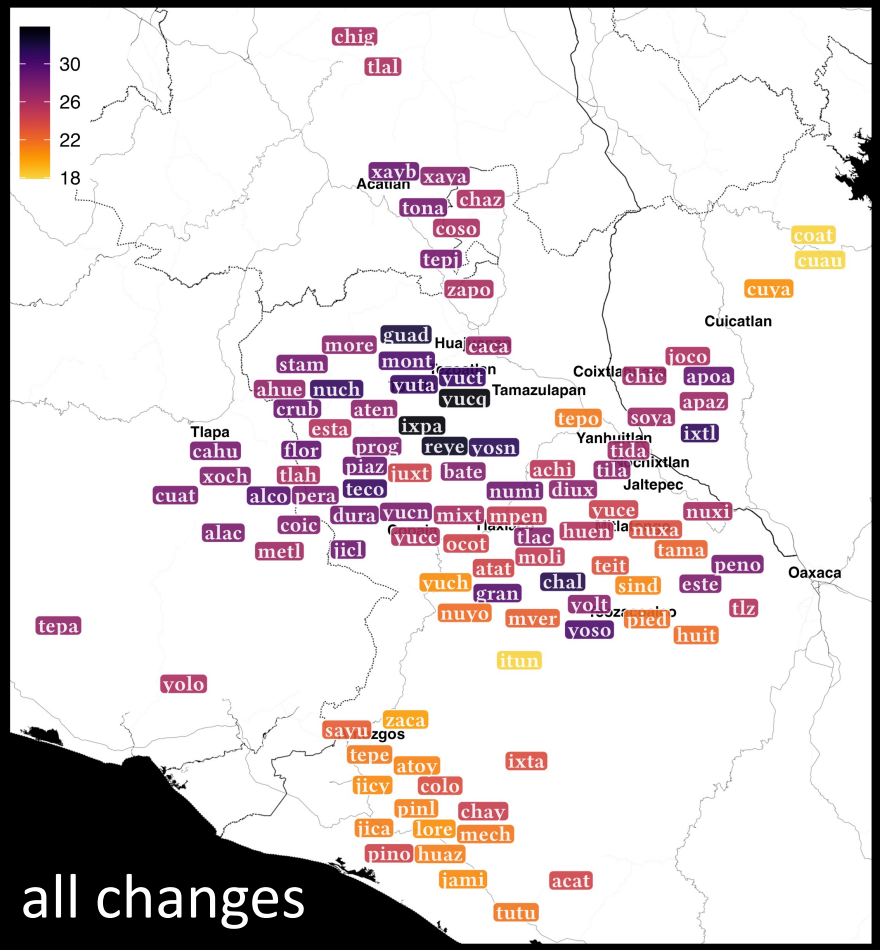
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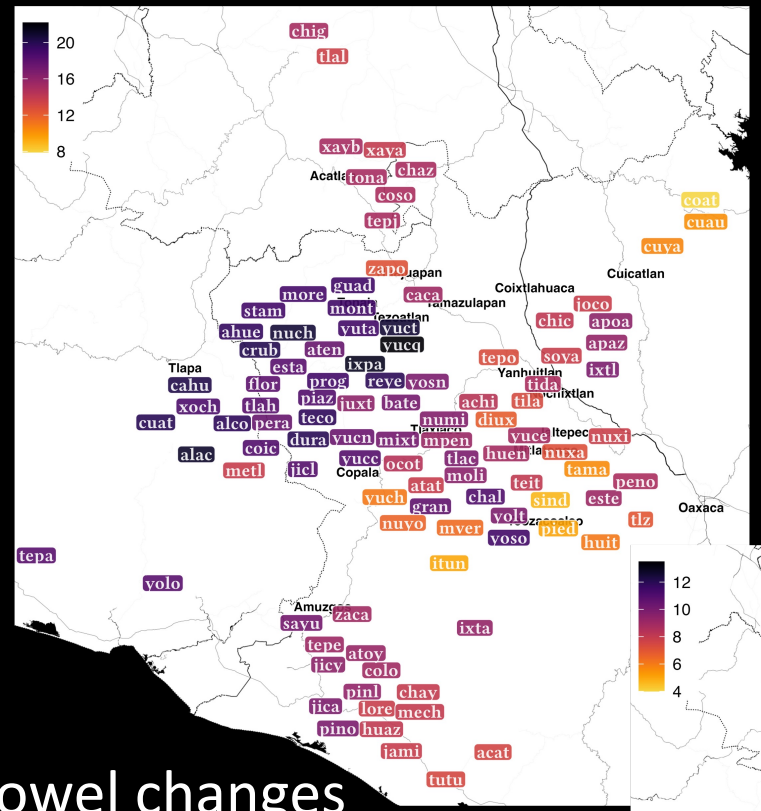
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Outlook:

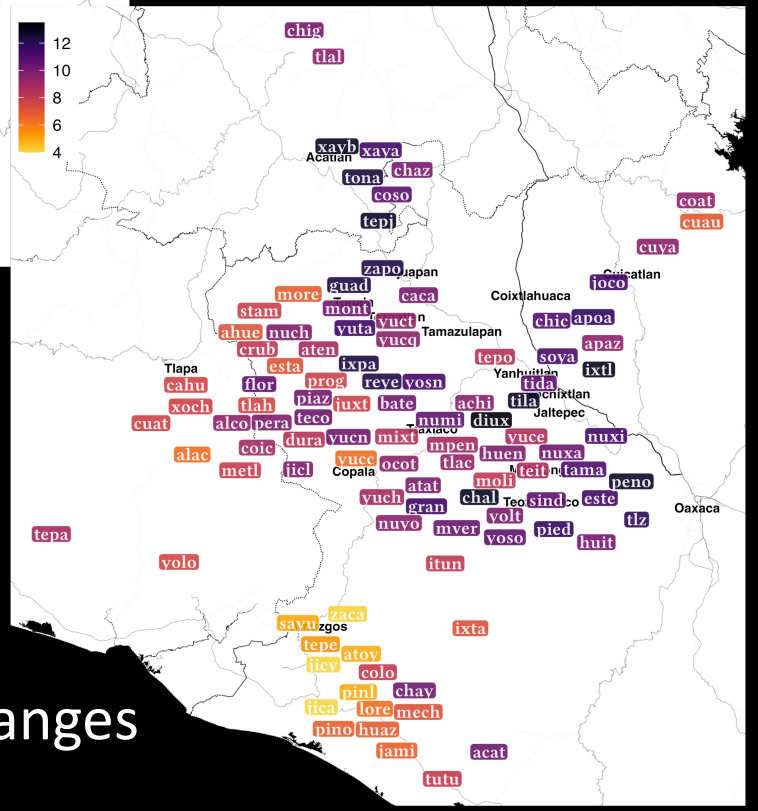
- making sound change & reconstruction more explicit, accessible, and re-usable
- application to other language families
- long run: typological data base of sound change but bottom-up



all changes

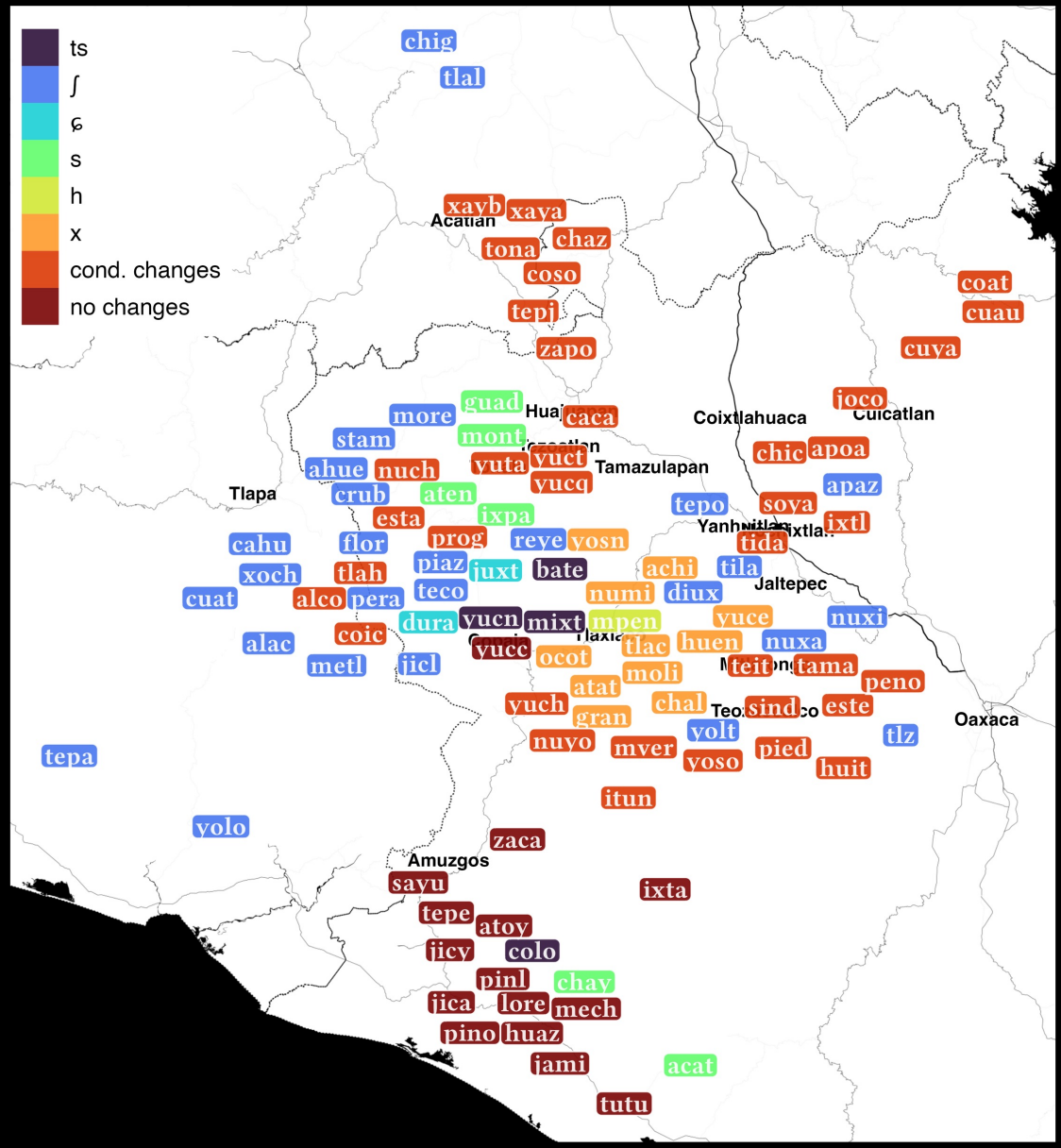


vowel changes



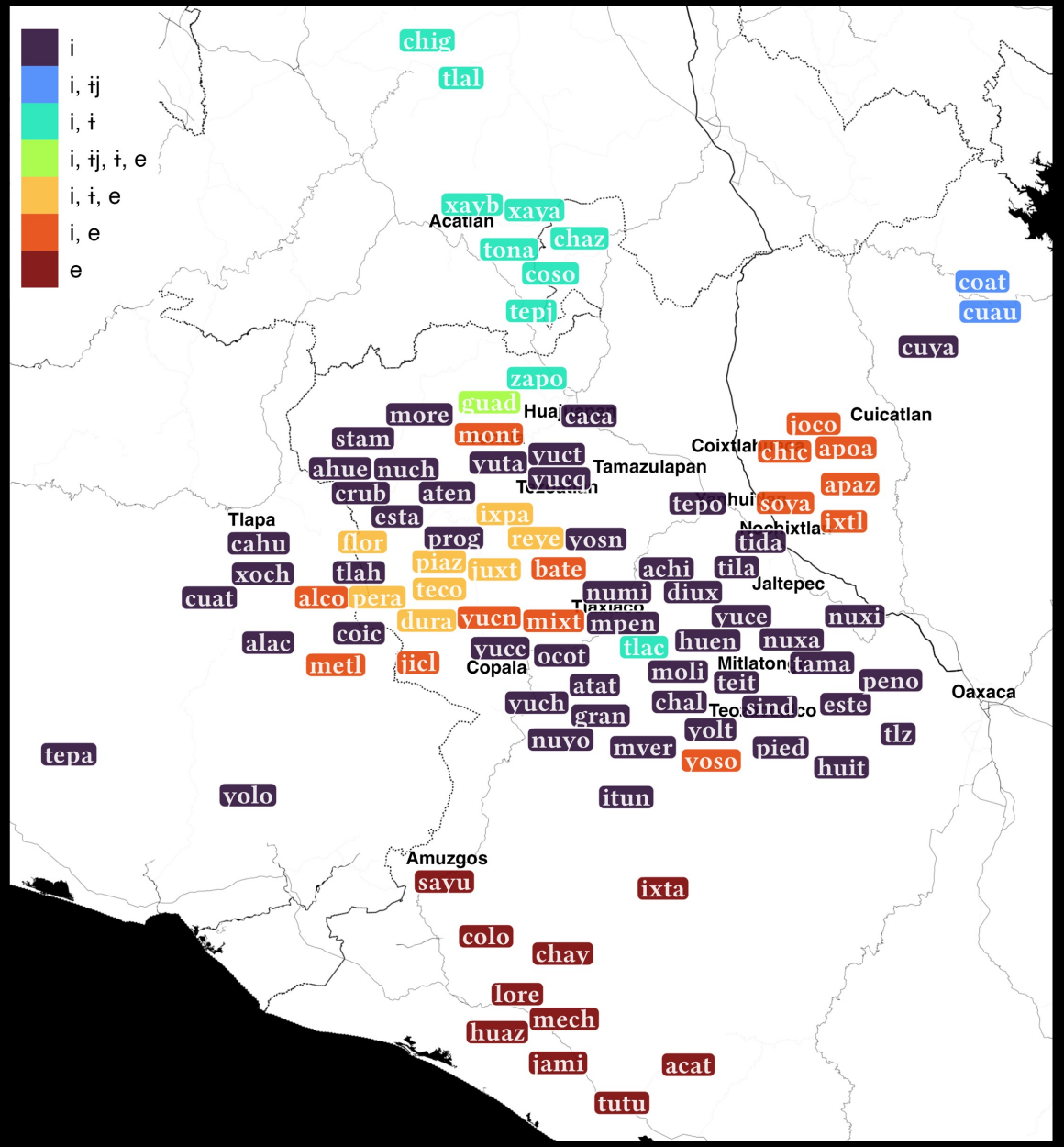
consonant changes

Global reflexes of PM *tʃ



Wave-like processes

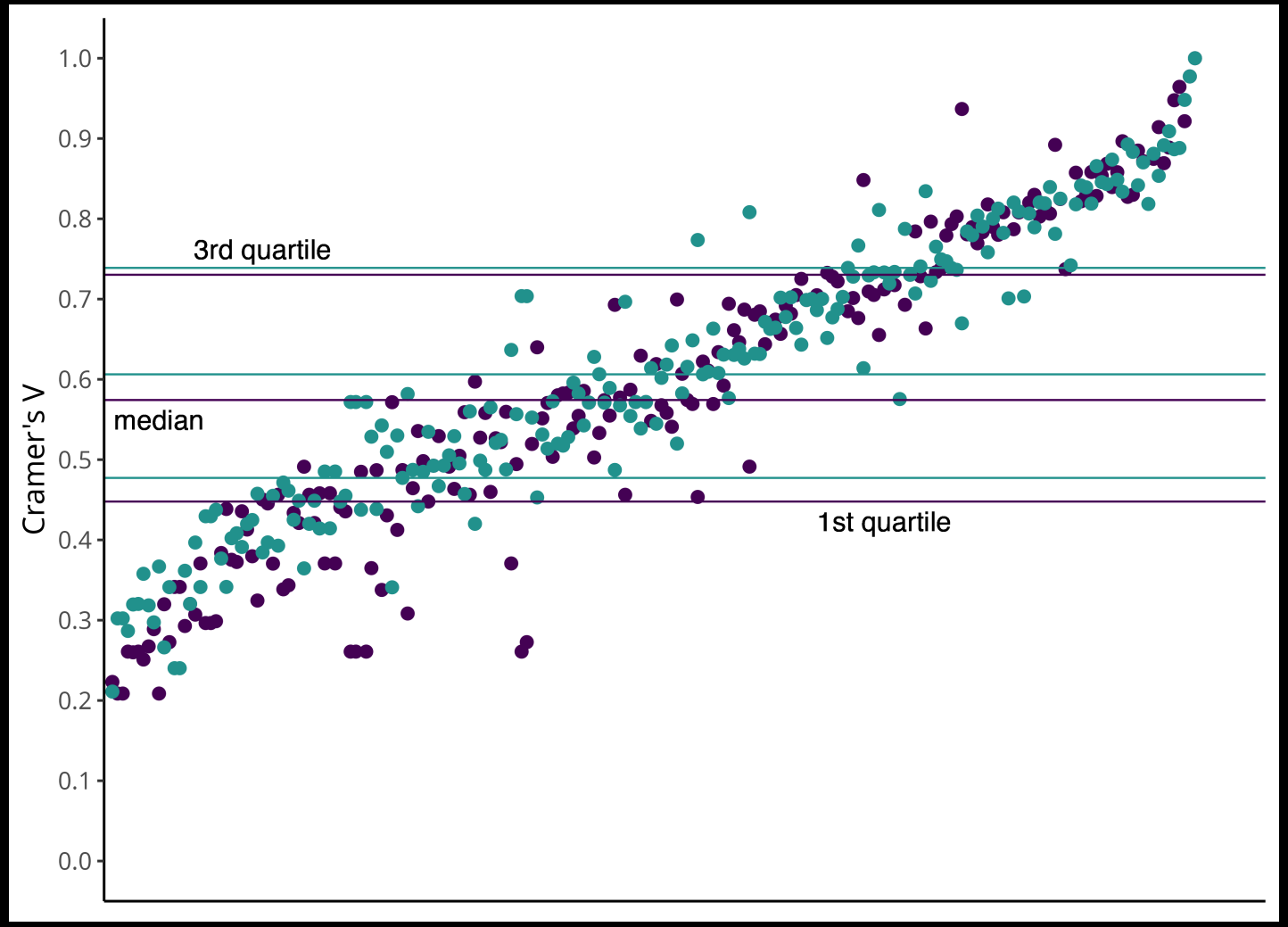
environments that condition palatalization of proto-Mixtec *t



Beyond trees and waves: Correlations

each point = sound change

teal = correlation with subgroups by Auderset et al. 2023
purple = correlation with dialect areas by Josserand 1983



Cramer's V (correlation measure)
> 0.6 strongly correlated
0.3-0.6 moderately correlated
< 0.3 weakly/not correlated